

John Harms

johnharms.netlify.app | jharms@uchicago.edu | Chicago, IL

EDUCATION

University of Chicago
Master of Science in Computational Analysis and Public Policy

Sep 2026 – Present

University of Wisconsin, Madison (GPA: 3.8/4.0)
Bachelor of Science in Mechanical Engineering, Physics Certificate

Aug 2021 – May 2026

- **Awards:** Dean's Honor List (x9), Madison G.L. Larson \$2500 Scholarship, Richard A. and Barbara A. Zevnik Memorial Scholarship \$2000
- **Relevant Coursework:** Applied Statistics, AI in Mechanical Engineering, MATLAB & Python Programming, Sociology of Race & Ethnicity, Renewable Energy Systems, Photovoltaic Arrays, Fluid Dynamics, Heat Transfer, Advanced CAD

QUANTITATIVE & DATA ANALYSIS EXPERIENCE

Machine Vision & Data Systems — 3M Company

Project Engineer Intern | Summer 2025

- Supported deployment of an industrial computer vision system, working with large-scale image data and feature extraction pipelines to detect manufacturing defects.
- Conducted data-driven root cause analysis linking visual patterns to upstream process conditions.
- Collaborated cross-functionally to translate quantitative findings into operational decisions, eliminating \$100k+ in annual waste.
- Proposed a machine-vision-based quality classification framework for fiber-optic connector defects, emphasizing interpretability and scalability.

ME 601 Final Project — Material Property Prediction Using Machine Learning

- Built a Python pipeline (scikit-learn, Matminer, DScribe) predicting elastic moduli for 1,181 DFT-computed crystal structures; gradient boosting on composition features reached $R^2 = 0.92$ on bulk modulus.
- Benchmarked four feature representations; SOAP structural descriptors (159,390 dimensions, PCA-reduced) scored $R^2 = 0.82$ alone and added nothing when combined with composition features.
- Traced the shortfall to information loss in descriptor averaging, and documented reliability limits across chemical subgroups.

Solar Array Feasibility Study — Bakke Recreation Center

- Modeled energy output, shading impacts, system cost, and ROI for campus solar installation using System Advisor Model (SAM) developed by NREL.
- Presented recommendations based on economic analysis, sustainability metrics, and system efficiency.

ENGINEERING EXPERIENCE

3M Company

Hutchinson, MN Summer 2025

Project Engineer Intern (Full-Time)

- Led an intern innovation project on process control of fiber optic connectors (EBOI), proposing a machine-vision approach to inspect and identify molding defects.
- Spearheaded multiple machine guarding and construction projects to improve operator safety and facility efficiency.

Sub-Zero Group, Inc.

Fitchburg, WI Jan 2024 – Aug 2024

Fabrication Co-op (Full-Time)

- Designed and upgraded multiple assembly tools and fixtures using SolidWorks and GD&T, improving ergonomics, versatility, and safety for operators while reducing costs of fixtures from \$15k to \$2k per fixture.
- Created contingency protocols and implemented reserve component storage, preventing costly production delays resulting in savings of \$600k within the first 8 months.
- Led projects from concept to implementation — sketching designs, sourcing raw materials, coordinating machining and assembly, and verifying functionality on the shop floor.
- Collaborated across engineering, supply chain, production, and management teams to align technical solutions with operational goals and ensure on-time delivery.

Energy Systems Lab

UW-Madison Jan 2023 – Sep 2023

Student Research Assistant (Paid)

- Supported energy research by creating SolidWorks models from drawings, machining prototype components, and assisting in test setups.
- Gained early exposure to molten salt based alternative energy systems, lab instrumentation, and CAD-to-fabrication workflows.

PROFESSIONAL EXPERIENCE

Wisconsin Aviation

Madison, WI Jun 2023 – Oct 2023

Line Service Technician

- Towed aircraft, fueled aircraft, and directed ground movement while communicating with pilots via radio in a high-risk, safety-critical environment.

Greenbush Bar

Madison, WI Jun 2023 – Dec 2023

Restaurant Server & Cook

- Managed customer service and coordinated front-of-house and kitchen communication in a fast-paced restaurant setting.

3M Company

Cottage Grove, MN Summer 2021

Production Operator

- Operated web handling equipment on large-scale manufacturing lines, maintaining quality consistency and safety standards.

LEADERSHIP & SERVICE

Peer mentoring in physics, CAD, thermodynamics, calculus

- Informally mentored classmates by breaking down complex engineering concepts into simple, visual explanations.
- Helped peers prepare for physics and mechanics exams by creating practice problems and guiding them through solution strategies.
- Earned a reputation among classmates and friends for being able to teach technical concepts clearly and patiently.

Engineers for a Sustainable World — Solar Dehydrator Build

- Collaborated with a small team to design and build a solar-powered food dehydrator for a local farmer, from concept through fabrication.
- Led construction sessions in the campus workshop, teaching peers safe tool use, woodworking techniques, and design interpretation.

CERTIFICATIONS & TECHNICAL SKILLS

Certifications

- Private Pilot License (2024), Instrument Rating (2025)

Software & Technical Tools

- SolidWorks (Advanced), MATLAB, Python, R, COMSOL, GD&T
- Industrial Tools: HALCON, Moldex3D
- Fabrication & Analysis: FEA Analysis, Topology Optimization, TPMS Modeling, Image-Based Defect Classification, Milling, Lathe Turning, CNC Collaboration, 3D Printing, Wood/Metal Fabrication

RESEARCH INTERESTS

Computational social science | Science, technology, and society (STS) | Data-driven policy analysis | Sociotechnical systems & institutional decision-making | Energy, infrastructure, and governance